

CHEM 101 Topic 8: Chemical Quantities — Answer Key

Section 1: Avogadro's Number and the Mole

Problem 1: 0.500 mol.

Problem 2: 4.52×10^{23} molecules.

Problem 3: 8.43×10^{24} atoms.

Section 2: Molar Mass and Mass-Mole Conversions

Problem 4: 101.11 g/mol.

Problem 5: 0.247 mol.

Problem 6: 11.90 g.

Section 3: Molarity

Problem 7: 2.00 M.

Problem 8: 0.800 M.

Problem 9: 18.02 g.

Section 4: Empirical and Molecular Formulas

Problem 10: Multiplier = 2.

Problem 11: $\text{C}_3\text{H}_6\text{O}_3$.

Problem 12: $\text{C}_9\text{H}_{15}\text{O}_3$.